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Coupling news sentiment with web browsing data predicts intra-day stock prices

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Abstract

The new digital revolution of big data is deeply changing our capability of understanding society and forecasting the outcome of many social and economic systems. Unfortunately, information can be very heterogeneous in the importance, relevance, and surprise it conveys, affecting severely the predictive power of semantic and statistical methods. Here we show that the aggregation of web users' behavior can be elicited to overcome this problem in a hard to predict complex system, namely the financial market. Specifically, we show that the combined use of sentiment analysis of news and browsing activity of users of *Yahoo! Finance* allows to forecast intra-day and daily price changes of a set of 100 highly capitalized US stocks traded in the period 2012-2013. Sentiment analysis or browsing activity when taken alone have very small or no predictive power. Conversely, when considering a *news signal* where in a given time interval we compute the average sentiment of the clicked news, weighted by the number of clicks, we show that for more the 50% of the companies such signal Granger-causes price returns. Our result indicates a "wisdom-of-the-crowd" effect that allows to exploit users' activity to identify and weigh properly the relevant and surprising news, enhancing considerably the forecasting power of the news sentiment.

1 Introduction

The recent technological revolution with widespread presence of computers, users and media connected by Internet has created an unprecedented situation of data deluge, changing dramatically the way in which we look at social and economic sciences. As people increasingly use the Internet for information such as business or political news, online activity has become a mirror of the collective consciousness, reflecting the interests, concerns, and intentions of the global population with respect to various economic, political, and cultural phenomena. Humans' interactions with technological systems are generating massive datasets documenting collective behaviour in a previously unimaginable fashion [1, 2]. By

properly dealing with such data collections, for instance representing them by means of network structures [3, 4], it is possible to extract relevant information about the evolution of the systems considered (i.e. trading [5], disease spreading [6, 7], political elections [8]).

A particularly interesting case of study is that of the financial markets. Markets can be seen as collective decision making systems, where exogenous (news) as well as endogenous (price movements) signals convey valuable information on the value of a company. Investors continuously monitor these signals in the attempt of forecasting future price movements. Because of their trading based on these signals, the information is incorporated into prices, as postulated by the Efficient Market Hypothesis [9]. Therefore the flow of news and data on the activity of investors can be used to forecast price movements. The literature on the relation between news and price movement is quite old and vast. In order to correlate news and price returns one needs to assess whether the former is conveying positive or negative information about a company, a particular sector or on the whole market. This is typically done with the sentiment analysis, often performed with dedicated semantic algorithms as described and reviewed in the Methods Section.

In this paper, we combine the information coming from the sentiment conveyed by public news with the browsing activity of the users of a finance specialized portal to forecast price returns at daily and intra-day time scale. To this aim we leverage a unique dataset consisting of a fragment of the log of *Yahoo! Finance*, containing the news articles displayed on the web site and the respective number of "clicks", i.e. the visualizations made by the users. Our analysis considers 100 highly capitalized US stocks in a one-year period between 2012 and 2013.

For each of these companies we build a signed time series of the sentiment expressed in the related news. The sentiment expressed in each article mentioning a company is weighted by the number of visualizations of the article. In our dataset each click action is associated with a timestamp recording the exact point in time when such action took place. Thus we are able to construct time series at the time resolution of the minute. To the best of our knowledge, this is the first time that an analysis like the one described in this paper is conducted at such intra-day granularity. The main idea behind this approach is that the sentiment analysis gives information on the news, while the browsing volume enable us to properly weigh news according to the attention received from the users.

We find that news on the same company are extremely heterogeneous in the number of clicks they receive, an indication of the huge difference in their importance and the interest these news generate on users. For 70% of the companies examined, there is a significant correlation between the browsing volumes of financial news related to the company, and its traded volumes or absolute price returns. More important, we show that for more than 50% of the companies (at hourly time scale), and for almost 40% (at daily time scale), the click weighted average sentiment time series Granger-cause price returns, indicating a rather large degree of predictability.

2 Data

2.1 Stocks considered

Our analysis is conducted on 100 highly capitalized stocks traded in the US equity markets, which we monitor during a period of one year between 2012 and 2013. The ticker list of the investigated stocks with a distinctive numerical company identifier follows: 1 KBH, 2 LEN, 3 COST, 4 DTV, 5 AMGN, 6 YUM, 7 UPS, 8 V, 9 AET, 10 GRPN, 11 ZNGA, 12 ABT, 13 LUV, 14 RTN, 15 HAL, 16 ATVI, 17 MRK, 18 GPS, 19 GILD, 20 LCC, 21 NKE, 22 MCD, 23 UNH, 24 DOW, 25 M, 26 CBS, 27 COP, 28 CHK, 29 CAT, 30 HON, 31 TWX, 32 AIG, 33 UAL, 34 TXN, 35 BIIB, 36 WAG, 37 PEP, 38 VMW, 39 KO, 40 QCOM, 41 ACN, 42 NOC, 43 DISH, 44 BBY, 45 HD, 46 PG, 47 JNJ, 48 AXP, 49 MAR, 50 TWC, 51 UTX, 52 MA, 53 BLK, 54 EBAY, 55 DAL, 56 NWSA, 57 MSCI, 58 LNKD, 59 TSLA, 60 CVX, 61 AA, 62 NYX, 63 JCP, 64 CMCSA, 65 NDAQ, 66 IT, 67 YHOO, 68 DIS, 69 SBUX, 70 PFE, 71 ORCL, 72 HPQ, 73 S, 74 LMT, 75 XOM, 76 IBM, 77 NFLX, 78 INTC, 79 CSCO, 80 GE, 81 WFC, 82 WMT, 83 AMZN, 84 VOD, 85 DELL, 86 F, 87 TRI, 88 GM, 89 FRT, 90 VZ, 91 FB, 92 BAC, 93 MS, 94 JPM, 95 C, 96 BA, 97 GS, 98 MSFT, 99 GOOG, 100 AAPL. The numerical identifiers are assigned according to the increasing order of the total number of published news in *Yahoo! Finance*.
We considered three main sources of data for these stocks:

2.2 Market data

The first source contains information on price returns and trading volume of the stock at the resolution of the minute. We consider different time scales of investigation, corresponding to 1, 10, 30, 65, and 130 minutes. The above values are chosen because they are sub-multiple of the trading day in the US markets (from 9:30 AM to 4:00 PM, corresponding to 390 minutes). For each time scale and each stock we extract the following time series:

- V, the traded volume in that interval of time,
- R, the logarithmic price return in the time scale,
- σ, the return absolute value, a simple proxy for the stock volatility.

The precise definition of these variables is given in the *Support Information*. Since trade volumes and absolute price returns are known to display a strong intra-day pattern, we de-seasonalize the corresponding time series (in the same SI Section we provide the details about this procedure). This procedure is necessary in order to avoid the detection of spurious correlation and Granger causality due to the presence of a predictable intra-day pattern.

2.3 News data

The second source of data consists of the news published on *Yahoo! Finance* together with the time series of the aggregated clicks made by the users browsing each page. *Yahoo! Finance* is the most popular web portal for news and data related to financial companies, offering news and information around stock quotes, stock exchange rates, corporate press releases, financial reports, and message boards for discussion. We analyze a portion of the web-site log, containing news articles displayed on the portal. The articles are manually tagged by a team of trained domain experts, who associate each article with the specific companies (e.g., Google, Yahoo!, Apple, Microsoft) or financial entities (e.g., market indexes, commodities, derivatives) that are mentioned in its text.

In order to automatically detect whether the article is conveying positive or negative news on the company, we perform a sentiment analysis. To obtain a sentiment score, we classify each article with *SentiStrength* [10], a state-of-the-art tool for extracting positive and negative sentiment from informal texts. The tool is based on a dictionary of "sentiment" words, which are manually picked by expert editors and annotated with a number indicating the amount of positivity or negativity expressed by them. The original dictionary of *SentiStrength* is not tailored to any specific knowledge or application domain, thus it is not the most proper choice to compute a *financial* sentiment. To solve this issue, we adapt the original dictionary by incorporating a list of sentiment keywords of special interest and significance for the financial domain [11]. Supported by previous research that studied stock price reaction to news headlines [12, 13, 14, 15], we simplify our data processing pipeline by performing the sentiment analysis computation on the title of each article, instead of using its whole content. The sentiment score is a simple sign $(-1, 0, +1)$ for each news depending on whether there are more positive or negative words in the title.

2.4 Browsing Data

Finally, in our analysis we use the information on the browsing volume, that is, the time series of "clicks" that the web users made on each article displayed on *Yahoo! Finance* to visualize its content. Given that the users' activity on this domain-specific portal proved to provide a clean signal of interest in financial stocks [16], we exploit it in this work to weight the sentiment of each article on a given financial company. Specifically, we use the number of clicks of an article as a proxy for the level of attention that users gave to that news. By aggregating over a time window the clicks on all the articles, even published earlier, that mention a particular company, it is possible to derive an estimation of the attention around that company.

In summary, for each time scale and for each stock, the variables we extract from the database are (see Support Information):

- C, the time series of the total number of clicks in a time window,

- S, the sum of the sentiment of all news related to each company,
- WS, the sum of the sentiment of all news weighted by the number of clicks.

The first quantity C is non negative and measures the level of attention in a given time interval for news about a specific company. The S variable is the usual sentiment indicator employed in numerous studies and provides the aggregated sentiment of the company specific news published in a given time interval. The most important and novel quantity is WS, which combines the two previous ones by assigning a sign to each click depending on the sentiment of the clicked news. As for the market variables, we remove the intra-day pattern from the click time series. In fact, both the publication of news [15] and the clicking activity of users [16] show a strong intra-day seasonality. These patterns are probably related to the way humans carry out their activities during the day (e.g. small activity during the lunch time, more hectic activity at the beginning or at the end of the business day).

3 Methods

3.1 Sentiment Analysis

Regarding the analysis of search-engine queries, some recent works [17, 18, 19, 20, 21] have studied the relation between the daily number of queries related to a particular stock and the trading volume over the same stock.

An incomplete list of contributions on the role of news includes studies investigating (i) the relation between exogenous news and price movements [12, 22, 23, 24], (ii) the correlation between high or low pessimism of media and high market trading volume [13]; (iii) the relation between the sentiment of news, earnings and return predictability [25, 26], (iv) the role of news in the trading action [15, 27, 28, 29]; (v) the role of macroeconomic news in the performance of stock returns [30], and (vi) the high-frequency market reaction to news [31].

For example, in a recent paper [18], related to ours, authors show that daily trading volumes of stocks traded at NASDAQ can be forecasted with the daily volumes of queries related to the same stocks. In another paper a similar analysis shows that an increase in queries predicts higher stock prices in the next two weeks [32]. As for social networks and micro-blogging platforms [33], Twitter data is becoming an increasingly popular choice for financial forecasting. For example some have investigated whether the daily number of tweets predicts SP 500 stock prices [34, 35]. A textual analysis approach to Twitter data can be found in other works [14, 36, 37, 38] where the authors find clear relations between mood indicators and Dow Jones Industrial Average. Some other authors have used news, Wikipedia data or search trends to predict market movements [22, 39, 40, 41].

There are two main critical aspects in the kind of analyses described above. First, the universe of all the search engine or social network users is probably too large and the fraction of users truly interested in finance is likely quite low, with the consequence that signals based on generic Internet users are very noisy. Second, as we will empirically show below, the universe of news considered is very heterogeneous in terms of their relevance as a signal of future price movement. For example, in a day there might be several positive but almost irrelevant news and only one negative but very important news on a company. Without weighting the relevance of the news, one can easily obtain the wrong signal. The intuition behind the current work is that the number of times a news is viewed by users is a measure of its importance as well as of the surprise it conveys. Moreover the users we consider are not generic, but are those who use one of the most important news and search portals for financial information, namely *Yahoo! Finance*.

3.2 Pearson Correlation

To overcome these limitations we collected for each stock and for each time scale a total of six time series, namely V, R, σ, C, S, and WS, and we study their dependence by making use of two tools. First, given two time series X_t and Y_t, we consider the Pearson's correlation coefficient

$$\rho(X,Y) = \frac{\langle X_t Y_t \rangle - \langle X_t \rangle \langle Y_t \rangle}{\sqrt{(\langle X_t^2 \rangle - \langle X_t \rangle^2)(\langle Y_t^2 \rangle - \langle Y_t \rangle^2)}} \tag{1}$$

where $\langle \cdot \rangle$ is the time average value. The correlation $\rho(X, Y)$ quantifies the linear contemporaneous dependence. In order to assess the statistical significance of the measured value we perform a statistical test of the null hypothesis that the correlation is zero by randomizing the time series.

3.3 Granger Causality

Our main goal is testing for the presence of statistical causality between the variables. To this end our second tool is the Granger causality test [42]. Granger's test is a common test used in time series analysis to determine if a time series X_t is useful in forecasting another time series Y_t. X_t is said to *Granger-cause* Y_t if Y_t can be better predicted using both the histories of X_t and Y_t rather than using only the history of Y_t. The Granger causality can be assessed by regressing Y_t on its own time-lagged values and on those of X_t. An F-test is then used to examine whether the null hypothesis that Y_t is not Granger-caused by X_t can be rejected with a given confidence level (in this paper we use a p-value of 5%).

4 Results

The most important aspect of our analysis is to test whether one can forecast financial variables, and more specifically price returns, by using the information on the browsing activity of the users. Namely if, by weighting the sentiment of the clicked news by the number of clicks each news receives, one can improve significantly the predictability of returns.

4.1 Heterogenous attention

The first observation is the extreme heterogeneity of the attention that users of *Yahoo! Finance* show towards the financial news of a given company. Figure 1 shows the complementary of the cumulative distribution function of the number of clicks per news concerning a given stock. There we show the curves for each of the top 10 stocks and for the aggregate of 100 stocks. In all cases the tail of the distribution is very well fit by a power law behavior [43] with a tail exponent very close to 1 (see Support Information) In fact, the mean exponent across all stocks is 1.15 ± 0.30 and restricting on the top 10 it is 0.99 ± 0.08. This indicates that there is a huge heterogeneity in the number of clicks a news receives and therefore in the importance users give to it. It is also a warning that not weighting properly the importance of the news can lead to overstate the importance of the many irrelevant news and to understate the importance of the few really important ones.

4.2 Synchronous correlation

In order to understand how the relation between financial and news variables depends on the time scale, we perform a synchronous correlation analysis. We remark that some variables (C, σ, V) are non negative, while others (R, S, WS) are signed. Not surprisingly thus, we find that the linear correlation between any one of the former and anyone of the latter is always very close to zero. For each of the 100 companies, we compute the Pearson's correlation coefficient ρ between the three sensible pairs made by one "news" time series and one "financial" time series. Figure 2 summarizes the results for the 65 min time series. The x axis lists the companies, uniquely identified by a number that provides the rank of the company in the order from the least to the most cited one (as measured by the absolute number of associated news). Thus, 1 corresponds to the company KBH with the least number of news, while 100 to the most cited AAPL. We label the y axis with the pairs (C, V), (C, σ), and (WS, R), while the color scale indicates the level of correlation. We compute the correlation sampling the original time series every 65 minutes, equalizing to zero those values whose significance does not reject the null hypothesis of zero correlation with 5% confidence. Figure 2 shows in general a positive but small level of correlation, similarly to the result obtained by Mao [14]. In the fourth row of Table 1 we report the percentage of the 100 companies for which we reject the null hypothesis of zero correlation at 5% confidence level.

4.3 Time scale

In order to investigate how the correlation changes with the time scale, in Table 1 we also show the percentage of rejection for the 1, 10, 30, and 130 minutes time scales. As a general comment, we observe

that the number of companies with a significant correlation becomes higher at finer time resolution. This is a known fact for market variables (e.g. volume and volatility), while we document it for the first time at intraday scale also for sentiment and browsing variables. As expected the correlations are stronger for unsigned rather than for signed variables. Nevertheless, we can state the presence of a significant relation between the attention given to news articles (signed on the basis of the sentiment expressed in them) related to a given stock and the evolution of the stock price. Remarkably, in the case of the sentiment time series a progressive increase of the number of companies can be observed, until we reach a hourly granularity, suggesting that this could be a characteristic time scale of the process.

4.4 Dynamics of attention

The time scale might in principle depend on the relevance of the news. As we have seen, not all news are equal in terms of the attention they receive from the users. To investigate this dependence, we study the dynamics of the number of clicks an article received after its publication. We compute the cumulative number of clicks received by a given news until a given minute after the publication. We perform this for all minutes in a week after the publication. We then normalize this cumulative time series by dividing it by the total amount of clicks received by the news. We construct ten groups of news based on the deciles of the total number of news they eventually receive, and we compute for each group the average cumulative sum of clicks. The result is shown in Figure 3. The inset reports for each decile the typical time scale of attention obtained by an exponential fit of the curves. Remarkably, the time scale of attention is an increasing function of the importance of the news (as measured by the total number of clicks). Irrelevant news are immediately recognized as such, while important news continue to receive attention well after their publication. In general, the time scale of the users' attention ranges between one and two hours after the publication, suggesting that this intraday time scale is probably the most appropriate to detect dependencies among financial variables and browsing activity.

4.5 Causality

The synchronous correlation is an important measure of dependence, but not necessarily a sign of causality. Thus we perform a causality analysis by applying Granger's test. We present the results of this analysis, for the 65-minute time horizon, in Figure 4. The x axis lists the companies as done in Figure 2, while the y axis labels the eight tests that we perform. Black cells correspond to rejection of the null hypothesis of no Granger causality, and the opposite for the white cells. When considering the non-negative variables (V, C, and σ) we observe strong causal relations. Specifically, in 65% of the cases the clicking activity causes the trading volume and in 69% of the cases it causes price volatility. The causality is very strong also in the opposite direction, i.e. volume and volatility cause click volume. This is probably due in part to a reaction of users to anomalously high activity in the market (in terms of volume and/or volatility), while in part it might be a statistical effect due to the fact that all the three variables are very autocorrelated in time, creating strong Granger causal relations in both directions.

We obtain the most interesting and unexpected results when considering the signed variables (R, S, and WS). All these variables are weakly serially autocorrelated. When we consider the sentiment of the news (S, without the clicks), we find that only in 4% of the cases S causes returns, and in 13% of the cases price return causes S. Especially the first value is expected under the null, since at 5% confidence level we expect 5% of false positive. This means that the simple sentiment of the news does not allow to forecast price returns at intraday (hourly) time scale. On the contrary, when we consider the clicks weighted by the sentiment of the news (WS), we find that in 53% of the cases it allows predicting returns and only in 19% of the cases the opposite occurs. In general, companies with more news have higher causality.

4.6 Weighting news by users' browsing behavior

These results show that, on a hourly time scale, the simple news sentiment time series alone (i.e. the one without browsing activity) is not able to predict the price returns; instead, if we add the information provided by the browsing activity, we are then able to properly weigh the news (and its sentiment) by the importance the users give to it by clicking the page. Thus, we find the interesting result that the

browsing activity combined with the sentiment analysis of the news increases significantly the level of prediction of price returns for the stocks.

4.7 Comparison with existing literature

Most of the existing studies on sentiment and predictability of returns focus on daily or longer time scale. In order to compare properly our result with the existing literature, we present in Table 2 the results of the above Granger tests on a daily time scale. Table 2 shows that, without the browsing activity, S causes R for 18% of the companies and R Granger-causes S in 9% of the cases. Thus there is now some predictability of sentiment, even if the number of companies is quite limited. This is consistent with the existing literature, which reports a weak daily predictability of returns by using sentiment. It is important to note that by adding the browsing activity we can double the number of companies for which there is predictability. In fact, WS Granger-causes R for 37% of the companies and 11% in the opposite direction.

5 Materials

5.1 Financial time series

We consider only trading hours (i.e. from 9:30 AM to 4:00 PM) and trading days (i.e. business days at the New York Stock Exchange), and we disregard the trading events that occur out of this time window. From the financial data we create three time series: the first one is the traded volume for each minute for each company, the second one is the logarithmic returns, and the last time series is the absolute value of the logarithmic returns.

1. **Trade volume (V) time series**. It consists of the traded volume of a given company at minute time scale. We build a time series at time scale t by summing the traded volumes v_τ at the smallest time scale τ (in our case one minute). Then, the total volume $\overline{v_t}$ can be defined as follows:

$$\overline{v_t} = \sum_{\tau > t-1}^{t} v_\tau \,. \tag{2}$$

   Traded volumes at intra-day time t are rescaled by a factor ζ_t^v, which is computed as the average, over all days, of the volume at time t normalized by the total daily volume. If $\overline{v_{d,t}}$ is the raw volume of day d and intra-day time t, we define the rescaled time series as

$$V_{d,t} = \frac{\overline{v_{d,t}}}{\zeta_t^v} \,, \tag{3}$$

   where

$$\zeta_t^v = \frac{1}{T} \sum_{d'} \frac{\overline{v_{d',t}}}{\Lambda_{d'}} \,, \tag{4}$$

   and

$$\Lambda_d = \sum_t \overline{v_{d,t}} \,, \tag{5}$$

   i.e. the total volume trades on day d, and T is the total number of trading days.

2. **Price return (R) time series**. We preliminary compute the logarithmic return, defined as

$$r_t = \log_{10}(\frac{p_t}{p_{t-1}}) \,, \tag{6}$$

   where p_t is the last recorded price in the interval $(t-1, t]$. Returns at intra-day time t are rescaled by a factor ζ_t^r, which is computed as the average, over all days, of absolute returns at time t rescaled by the average volatility. More precisely, if $r_{d,t}$ is the raw return of day d and intra-day time t, we define the rescaled time series as

$$R_{d,t} = \frac{r_{d,t}}{\zeta_t^r} \,, \tag{7}$$

where

$$\zeta_t^r = \frac{1}{T} \sum_{d'} \frac{|r_{d',t}|}{\Xi_{d'}} \,, \tag{8}$$

and

$$\Xi_d = \text{mean}(|r_{d,t}|) \,, \tag{9}$$

where we compute the mean value for all t in a day d.

3. **Volatility (σ) time series**. We employ as a simple proxy for the de-seasonalized intra-day volatility the absolute value of the rescaled return

$$\sigma_t = |R_t| \,. \tag{10}$$

Notice that the volatility σ_t is already de-seasonalized, because of the definition of R

5.2 Click and sentiment time series

Consistently with the financial time series, we consider only trading time and trading days. This implies that we neglect the clicks that occur out of this time window, because we are only interested in the click behaviour during the trading hours evolution of markets. From the click-through history of each news we create two time series: the first one is the total amount of clicks for each minute for each company, and the second one is the number of clicks multiplied by the sentiment score of the associated news.

1. **Click (C) time series**. Starting from the number of clicks of each news at the smallest time scale τ (in our case one minute), we build a time series at the time scale t by aggregating the number of clicks of all the news of a given company. So if we denote by N the number of news of a company, and by c_τ^i the number of clicks for news i at scale τ, the total volume $\overline{c}_t$ can be defined as

$$\overline{c_t} = \sum_{\tau > t-1}^{t} \sum_{i=1}^{N} c_\tau^i \,. \tag{11}$$

The news that are not viewed in the time interval have zero clicks. We filter out the intra-day pattern from clicks by means of a simple methodology. Clicks at intra-day time t are rescaled by a factor ζ_t^c, which is computed as the average, over all days, of the click volume at time t normalized by the total number of daily clicks. More precisely, if $\overline{c_{d,t}}$ is the raw click volume of day d and intra-day time t, we define the time series of rescaled clicks as:

$$C_{d,t} = \frac{\overline{c}_{d,t}}{\zeta_t^c} \,, \tag{12}$$

where

$$\zeta_t^c = \frac{1}{T} \sum_{d'} \frac{\overline{c}_{d',t}}{\Gamma_{d'}} \,, \tag{13}$$

and

$$\Gamma_d = \sum_t \overline{c}_{d,t} \,, \tag{14}$$

with Γ_d the total number of clicks in a day.

2. **Sentiment (S) time series**. To construct this time series we consider the sentiment of the headlines of the news. With the same notation previously used

$$S_t = \sum_{\tau > t-1}^{t} \sum_{i=1}^{N} s_\tau^i \,, \tag{15}$$

where s_τ^i is the sign $(-1, 0, 1)$ of the sentiment of a news headline published at time τ.

3. **Weighted Sentiment (WS) time series**. We multiply the click count of each news by its sentiment score. With the same notation of the click time series, we have:

$$\overline{WS}_t = \sum_{\tau>t-1}^{t} \sum_{i=1}^{N} c_\tau^i s_\tau^i \,. \tag{16}$$

This way, we weight the sign of every news by the level of attention that the news has received. We remark that for each trading day we consider all the clicks of all the news clicked in that day, not only the news that have been released in that day. Then, we define the weighted sentiment as

$$WS_t = \text{sign}(\overline{WS}_t) C_t \,. \tag{17}$$

5.3 Estimation of power law exponents

We estimate the power law tail exponent of the distribution of the number of clicks per news by employing the R package PoweRlaw developed and maintained by Colin Gillespie, and described in [45].

6 Discussion

The semantic analysis of the news on a specific company is known to have a small predictive power on the future price movements. At the light of our findings, we argue that this effect could be related to the distribution in the attention that the news receive, as clearly emerge in Figure 1: its scale-free behaviour reflects the extreme heterogeneity in the information they convey and the surprise they generate in the readers.

We show that by adding the clicking activity of the web users, we can increase dramatically the predictive power of the news for the price returns. This occurs because the time series built with only the sentiment of the news gives the same weight to all the news. In this way even irrelevant news are considered, adding noise to the sentiment time series and reducing the predictive power of the signal. Adding the browsing activity means giving a meaningful weight to each news according to its importance, as measured by the attention it receives by the users, and this largely improves the predictability of returns.

The approach to collective evaluation that we proposed in this paper can be useful in many other non financial contexts, since the overflow of information is a common aspect in our lives. In the financial domain, a natural extension of the present work concerns market instabilities and crashes. The analysis presented here is in fact unconditional, i.e. it does not target large price movements or, more generally, abnormal returns. From our societal perspective it would be extremely valuable to have a collective evaluation system, like the one presented here, capable of sifting the relevant information from the pool of data, news, blogs, etc, and to provide early warning indicators of large price movements. Since we have shown that predictability exists also at intraday time scale, this approach could be also used for real-time indicators, as well as for high-frequency instabilities and systemic price cojumps [44], which are becoming increasingly more frequent in our highly automated financial world.

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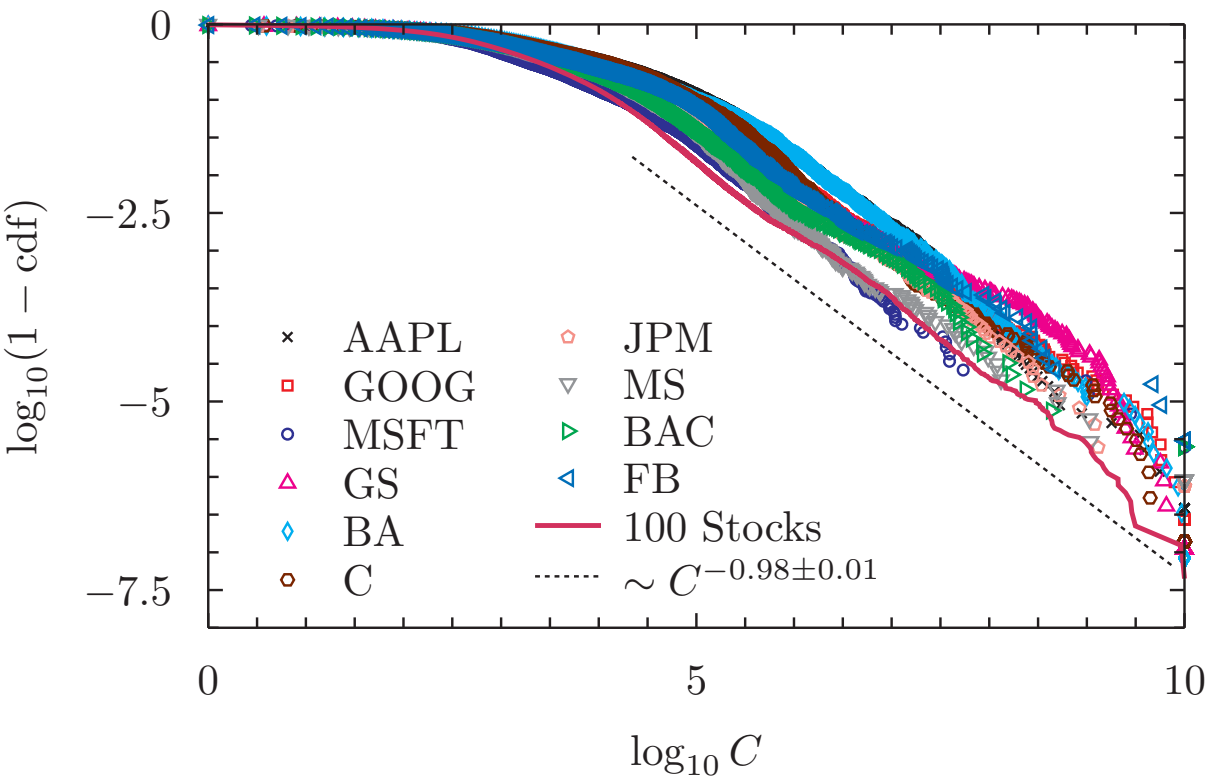

Figure 1: Complementary of the cumulative distribution function of the number of clicks a news receives for the ten assets with the largest number of news and the aggregate portfolio of 100 stocks. Both coordinates have been rescaled by a common factor preserving the power law scaling of the right tail and normalizing the maximum number of clicks to the value 10^{10}. The dotted line corresponds to a power law with tail exponent fitted from the portfolio time series. We provide details about the standard error and the complete list of tail exponents for all the companies in Support Information.

Table 1: Percentage of companies for which we reject the null hypothesis of zero correlation at 5% confidence level.

Time interval (minutes)	$\rho(WS, R)$	$\rho(C, \sigma)$	$\rho(C, V)$
1	6	77	78
10	8	48	69
30	10	32	62
65	13	30	57
130	8	29	52

Table 2: Number of companies for which we reject the null hypothesis of no Granger causality at 5% confidence level.

Causality relation	Hourly scale	Daily scale
$S \to R$	4	18
$R \to S$	13	9
$R \to WS$	19	11
$WS \to R$	53	37
$V \to C$	100	97
$C \to V$	65	52
$C \to \sigma$	69	52
$\sigma \to C$	96	16

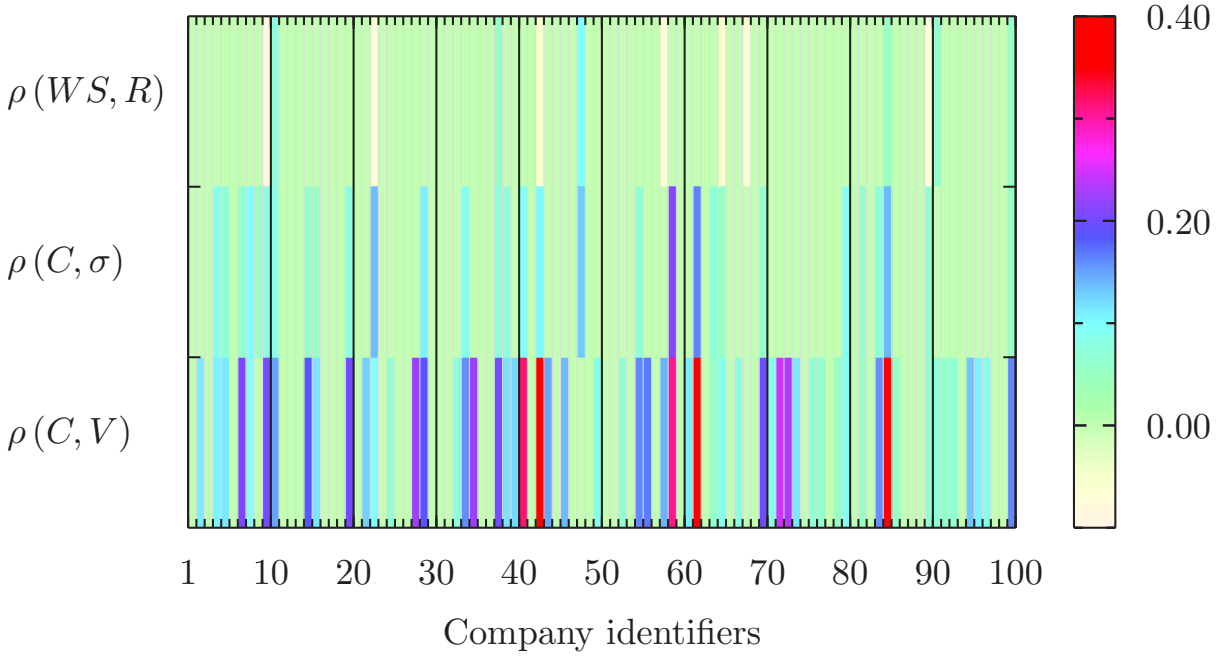

Figure 2: Pearson's correlation coefficients for the de-seasonalized time series of all the 100 companies at hourly scale. The x axis reports the list of companies identified by a unique number, as detailed in the main text. Among the several possibilities, we consider only three couples and the color scale corresponds to the level of correlation. We plot those values for which we reject the null of zero correlation at 5% significance level and equalize non significant values to zero (light green color).

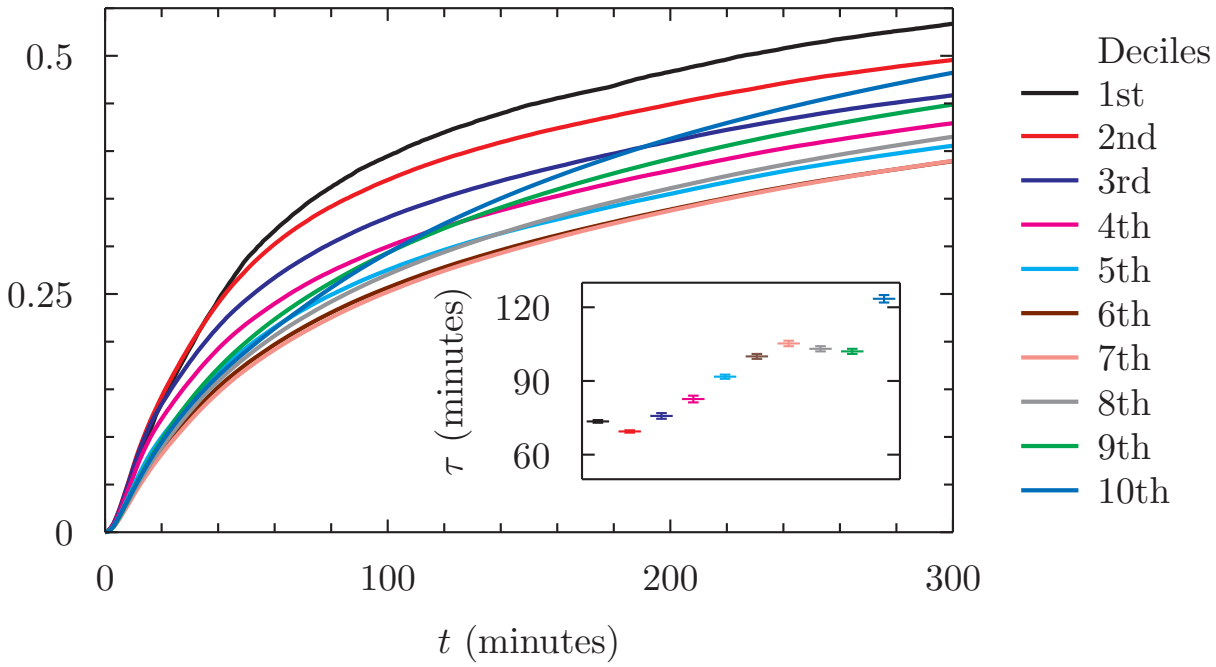

Figure 3: Time evolution of the cumulative number of clicks per news in a time interval of five hours after the publication. We normalize the cumulated amount by a constant which corresponds to the total number of clicks received by a single news during the first week after publication. The news are grouped in deciles according to the total number of clicks they have received until October 2013 and the curves represent average values. Inset: estimated values and standard errors of the attention time scale obtained by an exponential fit of the decile curves.

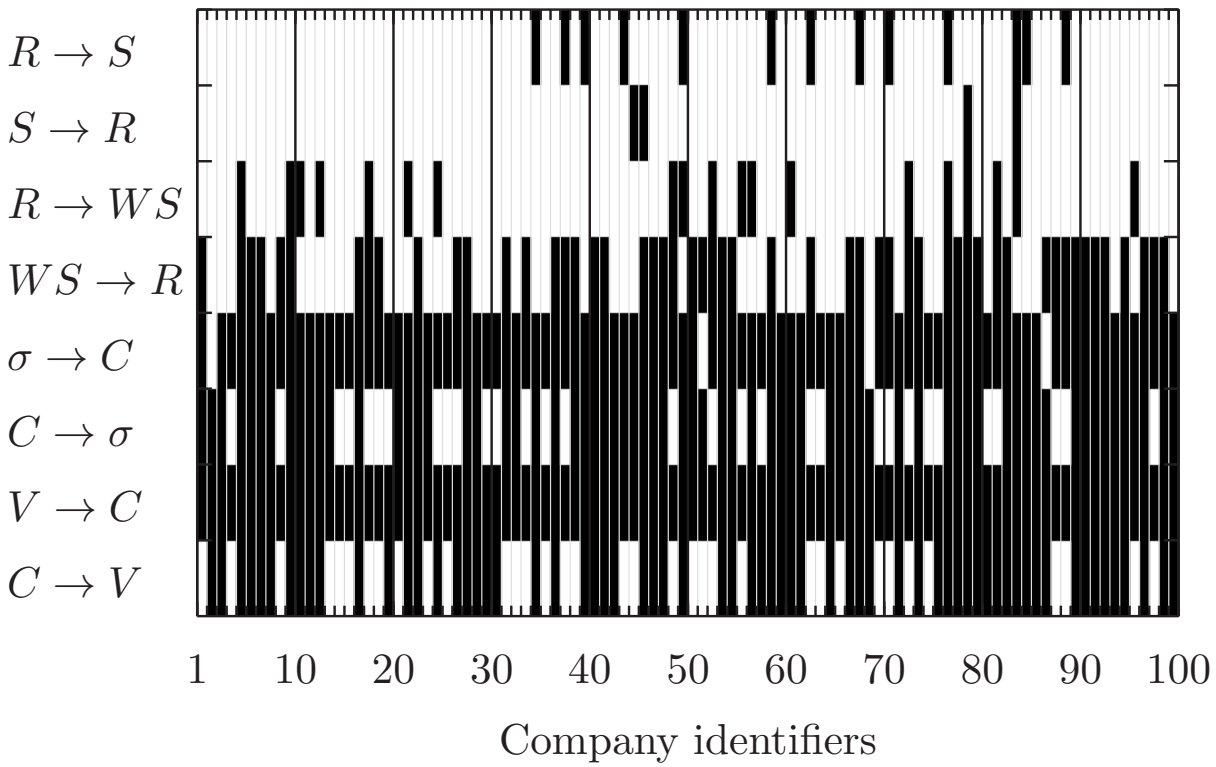

Figure 4: Granger Causality tests at hourly scale between de-seasonalized time series (x axis as in figure 2). The white cells correspond to tests for which we do not reject the null hypothesis of no Granger causality at 5% significance level. A black cell corresponds to a statistically significant Granger causality.

Table 3: Tail exponent α and lower bound $x_{\min} > 0$ with standard errors ϵ_α and $\epsilon_{x_{\min}}$ estimated from 100 highly capitalized stocks traded in the US equity markets. Following the procedure detailed in [45], we estimate the probability distribution associated with integer numbers larger than $x_{\min}$ whose expression reads to $p(x) = x^{-1-\alpha}/\zeta(1+\alpha, x_{\min})$. The normalizing constant corresponds to the Hurwitz zeta function ζ. We report in bold the top 10 assets, as measured by the absolute number of associated news.

	α	ϵ_α	$x_{\min}$	$\epsilon_{x_{\min}}$		α	ϵ_α	$x_{\min}$	$\epsilon_{x_{\min}}$
AA	1.05	0.09	482	114	**JPM**	**0.99**	**0.04**	**665**	**205**
AAPL	**0.97**	**0.05**	**2881**	**1451**	KBH	1.33	0.20	402	98
ABT	1.39	0.33	530	261	KO	0.96	0.09	565	242
ACN	1.84	0.37	1074	313	LCC	0.95	0.08	722	371
AET	1.21	0.13	261	54	LEN	0.86	0.19	256	197
AIG	1.16	0.13	854	193	LMT	1.41	0.14	808	159
AMGN	1.60	0.33	1596	445	LNKD	0.90	0.13	735	297
AMZN	1.06	0.05	969	206	LUV	0.82	0.10	617	282
ATVI	1.47	0.23	863	192	MA	0.97	0.13	335	213
AXP	0.89	0.07	447	93	MAR	1.34	0.17	318	90
BAC	**0.95**	**0.05**	**807**	**200**	MCD	0.71	0.06	462	305
BA	**1.01**	**0.05**	**802**	**250**	M	0.86	0.09	384	200
BBY	0.74	0.13	957	526	MRK	1.53	0.20	966	213
BIIB	1.88	0.41	1081	251	MSCI	1.12	0.09	205	62
BLK	1.20	0.12	547	122	**MS**	**1.15**	**0.09**	**615**	**457**
CAT	1.14	0.11	834	224	**MSFT**	**1.06**	**0.05**	**2247**	**507**
CBS	0.82	0.07	169	75	NDAQ	1.24	0.08	213	42
C	**1.10**	**0.06**	**627**	**256**	NFLX	0.94	0.07	613	141
CHK	1.72	0.18	1479	215	NKE	1.09	0.14	566	143
CMCSA	1.41	0.24	1186	357	NOC	1.46	0.22	894	228
COP	1.76	0.24	1708	303	NWSA	0.89	0.06	234	117
COST	0.79	0.10	448	372	NYX	0.98	0.18	176	192
CSCO	1.32	0.11	1290	271	ORCL	0.95	0.20	523	422
CVX	1.37	0.15	782	180	PEP	0.92	0.13	565	220
DAL	1.01	0.12	959	220	PFE	1.54	0.14	1306	250
DELL	1.00	0.07	944	246	PG	1.02	0.11	809	327
DIS	0.72	0.04	283	198	QCOM	1.44	0.23	1748	363
DISH	1.05	0.17	515	391	RTN	1.83	0.47	1379	425
DOW	1.21	0.13	818	313	SBUX	0.80	0.07	453	236
DTV	1.11	0.13	438	132	S	1.09	0.14	684	424
EBAY	0.91	0.11	1612	459	TRI	0.89	0.04	122	62
FB	**0.95**	**0.08**	**1211**	**1077**	TSLA	0.98	0.11	3024	850
F	0.96	0.05	1188	254	TWC	1.06	0.09	403	176
FRT	1.19	0.08	412	87	TWX	0.84	0.08	260	119
GE	0.98	0.08	587	236	TXN	1.55	0.42	1060	475
GILD	1.87	0.43	1300	294	UAL	1.00	0.12	1153	659
GM	0.84	0.05	863	296	UNH	1.21	0.15	378	147
GOOG	**0.92**	**0.03**	**1786**	**450**	UPS	1.93	0.49	2499	721
GPS	1.28	0.17	390	158	UTX	1.49	0.20	733	209
GRPN	0.93	0.13	644	193	V	1.17	0.13	505	243
GS	**0.89**	**0.05**	**462**	**143**	VMW	1.50	0.45	1003	547
HAL	1.56	0.18	880	213	VOD	1.87	0.42	1850	711
HD	0.79	0.07	456	169	VZ	1.11	0.10	1001	359
HON	1.18	0.17	632	296	WAG	1.06	0.23	512	322
HPQ	1.13	0.11	786	274	WFC	0.95	0.05	874	271
IBM	1.11	0.11	1014	283	WMT	0.67	0.06	2073	517
INTC	1.35	0.32	1777	738	XOM	1.29	0.12	802	147
IT	1.32	0.13	410	98	YHOO	1.15	0.12	2546	605
JCP	0.75	0.07	2827	959	YUM	0.90	0.10	483	225
JNJ	1.51	0.19	1058	174	ZNGA	1.38	0.16	1578	338